MSG No. 136.504 P_c4_2/mnm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,6,9,14,18,19,26,27]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,10,11,17,22,25,30]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,12,13,23,24,31,32]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[7,8,15,16,20,21,28,29]`

Table 2: Wyckoff site: `4b`, site symmetry: `4'/m'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,6,10,11,18,19,25,30]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[2,3,9,14,17,22,26,27]`
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4,5,15,16,23,24,28,29]`
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[7,8,12,13,20,21,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,15,16,23,24,25,30]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,12,13,20,21,26,27]`
3	$\left[0, 0, \frac{1}{4}\right]$	`[4,5,10,11,18,19,28,29]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[7,8,9,14,17,22,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,6,7,8,9,14,15,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,22,23,24,25,30,31,32]`
4	$[0, 0, 0]$	`[18,19,20,21,26,27,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,6,18,19]`
2	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[2,3,17,22]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[4,5,23,24]`
4	$\left[0, \frac{1}{2}, -z\right]$	`[7,8,20,21]`
5	$\left[\frac{1}{2}, 0, -z\right]$	`[9,14,26,27]`
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[10,11,25,30]`
7	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[12,13,31,32]`

8	$\left[0,\ \frac{1}{2},\ z\right]$	`[15,16,28,29]`

Table 6: Wyckoff site: `8f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,6,15,16]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,10,11]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[7,8,9,14]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,22,31,32]`
6	$[0,\ 0,\ z]$	`[18,19,28,29]`
7	$[0,\ 0,\ -z]$	`[20,21,26,27]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[23,24,25,30]`

Table 7: Wyckoff site: `8g`, site symmetry: `m.2m`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7,14,16]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,5,11,12]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,4,10,13]`
4	$[-x,\ -x,\ 0]$	`[6,8,9,15]`
5	$\left[x,\ x,\ \frac{1}{2}\right]$	`[17,23,30,32]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[18,21,27,28]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[19,20,26,29]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[22,24,25,31]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[2,12,21,27]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4,10,19,29]`
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[5,11,18,28]`
6	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[6,15,24,25]`
7	$\left[x,\ x,\ \frac{3}{4}\right]$	`[7,14,17,32]`
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[8,9,22,31]`

Table 9: Wyckoff site: `16i`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,11]`
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,10]`

continued ...

Table 9

No.	position	mapping
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[4,13]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$[y,\ x,\ 0]$	[7,16]
8	$[-y,\ -x,\ 0]$	[8,15]
9	$\left[x,\ y,\ \frac{1}{2}\right]$	[17,30]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ 0\right]$	[18,27]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[19,26]
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[20,29]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[21,28]
14	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[22,25]
15	$\left[y,\ x,\ \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[24,31]

Table 10: Wyckoff site: 16j, site symmetry: m'..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,30]
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[2,27]
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,26]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[4,29]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[5,28]
6	$\left[-x,\ -y,\ \frac{1}{4}\right]$	[6,25]
7	$\left[y,\ x,\ \frac{3}{4}\right]$	[7,32]
8	$\left[-y,\ -x,\ \frac{3}{4}\right]$	[8,31]
9	$\left[-x,\ -y,\ \frac{3}{4}\right]$	[9,22]
10	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[10,19]
11	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[11,18]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[12,21]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[13,20]
14	$\left[x,\ y,\ \frac{3}{4}\right]$	[14,17]
15	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[15,24]
16	$\left[y,\ x,\ \frac{1}{4}\right]$	[16,23]

Table 11: Wyckoff site: 16k, site symmetry: ..m

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[2,12]
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3,13]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[4,10]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5,11]
6	$[-x,\ -x,\ z]$	[6,15]
7	$[x,\ x,\ -z]$	[7,14]

continued ...

Table 11

No.	position	mapping
8	$[-x,\ -x,\ -z]$	[8,9]
9	$\left[x,\ x,\ z+\frac{1}{2}\right]$	[17,32]
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[18,28]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[19,29]
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[20,26]
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	[21,27]
14	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	[22,31]
15	$\left[x,\ x,\ \frac{1}{2}-z\right]$	[23,30]
16	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	[24,25]

Table 12: Wyckoff site: 321, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[2]
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[13]
14	$[x,\ y,\ -z]$	[14]
15	$[-y,\ -x,\ z]$	[15]
16	$[y,\ x,\ z]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[18]
19	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[19]
20	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[20]
21	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[21]
22	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[24]
25	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[28]
29	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[29]
30	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[30]
31	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[32]